

Unit 7b: Extreme Weather; Weather Versus Climate

Reading: pages 105–114 · 20 points · 15 minutes · Closed book

Name _____

Date _____

Score / 20

Answer every question. For multiple choice, circle the single best answer. Write short answers in complete sentences.

Part A · Multiple Choice

6 questions × 2 points = 12 points

1. During which stage of a thunderstorm's life cycle do updrafts and downdrafts exist side by side, with the storm at its most intense?
 - a) Mature stage
 - b) Dissipating stage
 - c) Entrainment stage
 - d) Cumulus stage
2. Of all the moisture held in a thunderstorm cloud, roughly what share actually reaches the ground as precipitation?
 - a) 20%
 - b) 50%
 - c) 80%
 - d) 95%
3. The intensity of a tornado is measured on the
 - a) Saffir-Simpson Scale
 - b) Fujita Scale
 - c) Richter Scale
 - d) Köppen Scale
4. Hurricanes arise in the tropical latitudes when sea surface temperatures reach at least
 - a) 18°C (64°F)
 - b) 24°C (75°F)
 - c) 28°C (82°F)
 - d) 35°C (95°F)
5. Within a hurricane, the strongest winds, tallest clouds, lowest atmospheric pressure, and most intense precipitation are found in the
 - a) eye
 - b) eyewall
 - c) spiral rain bands
 - d) storm surge

6. In the Köppen system, a climate with extremely cold winters and a warmest-month average temperature below 10°C (50°F) is classified as
- a) Tropical (A)
 - b) Dry (B)
 - c) Continental (D)
 - d) Polar (E)

Part B · True or False

2 questions × 1 point = 2 points

7. Meteorologists typically use data spanning about 30 years to define a region's climate.
TRUE / FALSE
8. A mid-latitude cyclone rotates clockwise in the Northern Hemisphere.
TRUE / FALSE

Part C · Short Answer

2 questions × 3 points = 6 points

9. Explain the difference between *weather* and *climate*, and name what a meteorologist studies compared with what a climate scientist studies.

10. What is a *storm surge*, and why is it so devastating along low-lying coastlines?

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